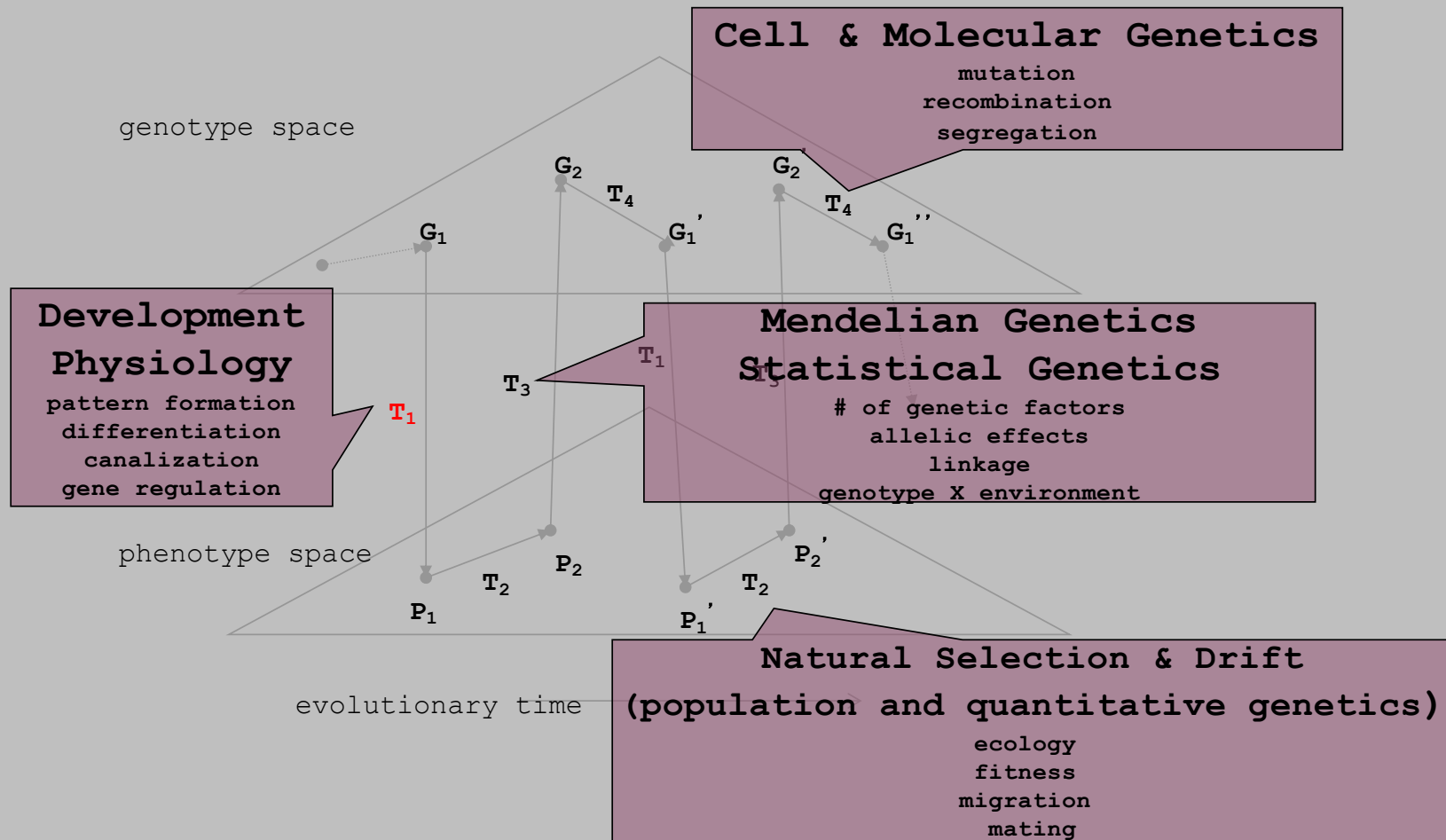




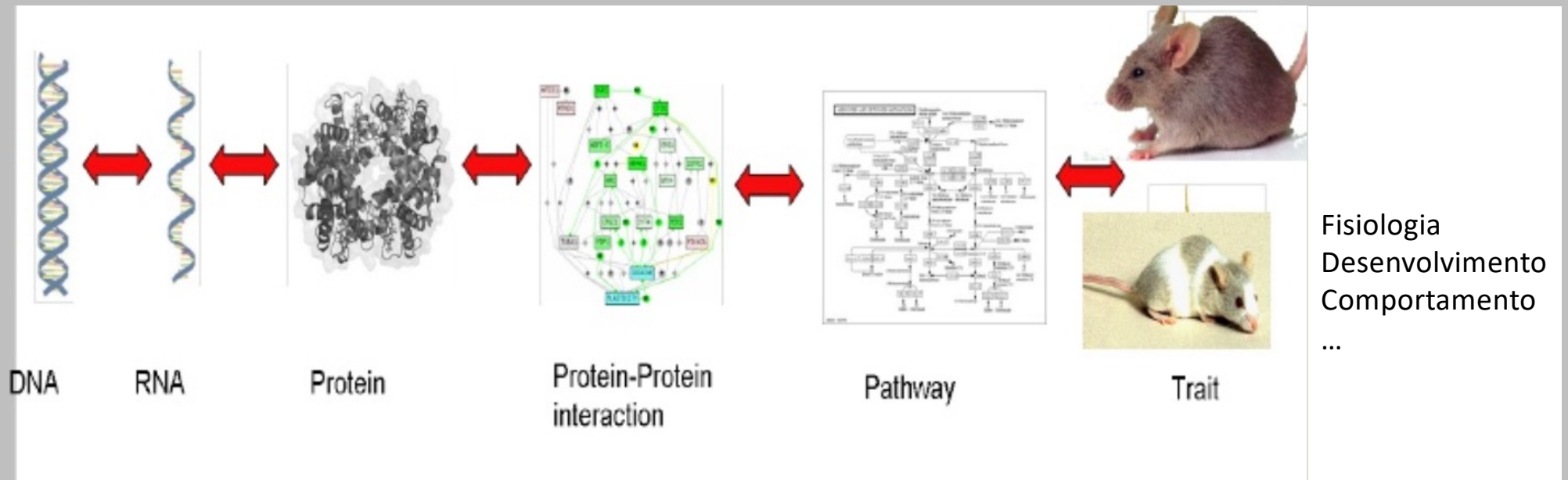
$$\text{Phenotype} = \text{Genetics} + \text{Environment} + \text{GxE}$$

$$(V_p = V_g + V_e + V_{g \times e})$$

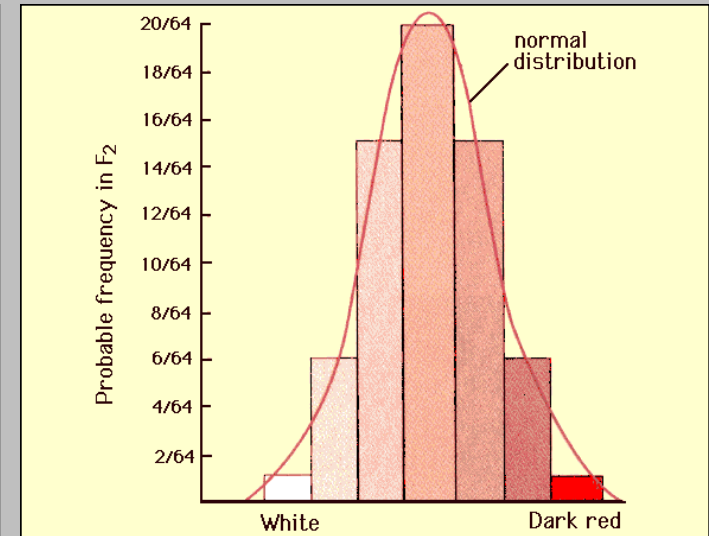
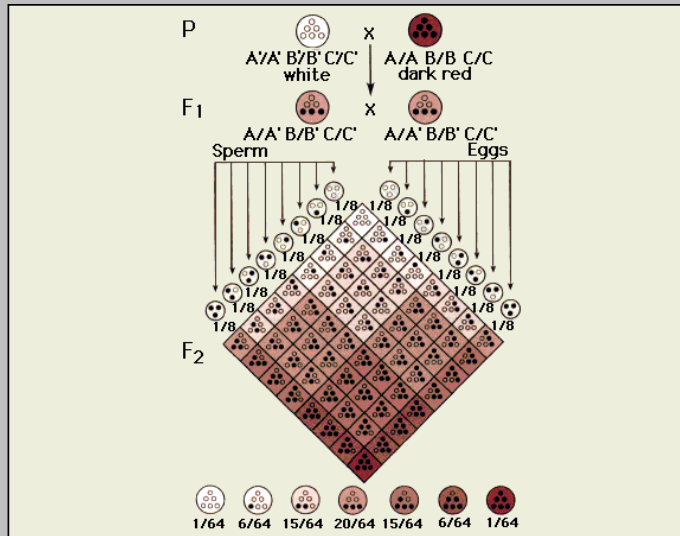
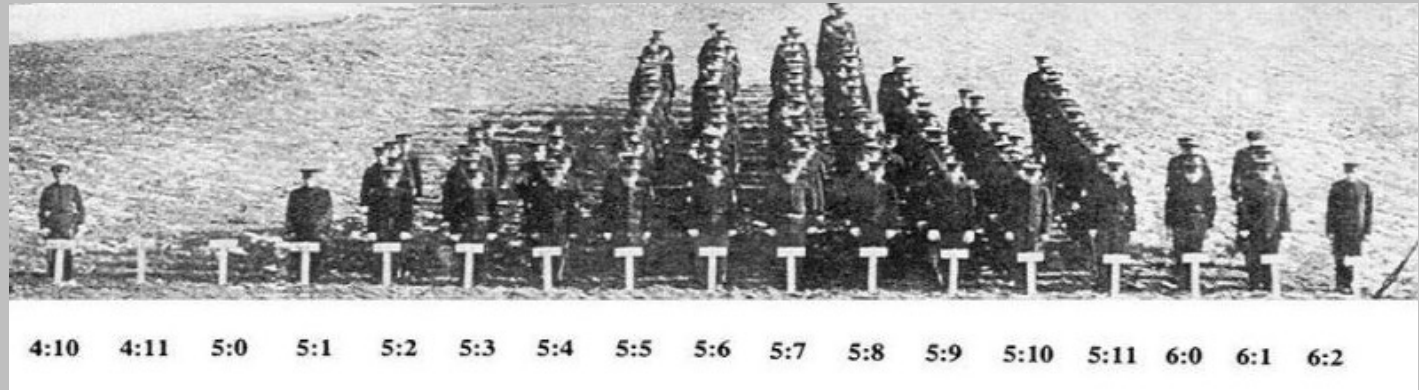
Fenótipo pode ser definido a qualquer nível de organização biológica

Fenótipo:

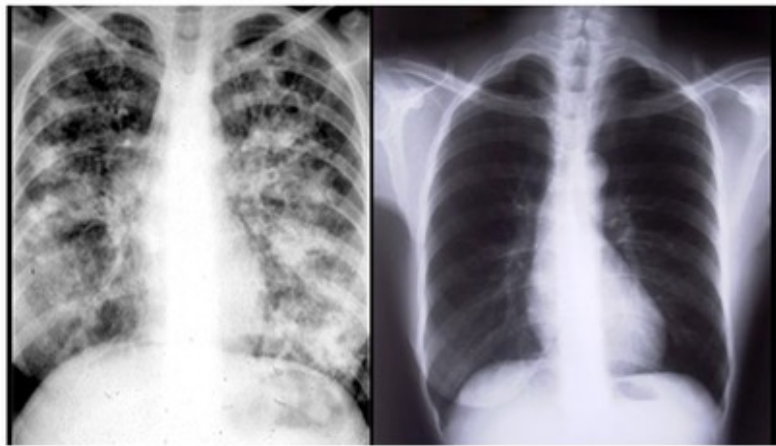
manifestação observável e quantificável dessa identidade



VARIAÇÃO



Fibrose cística e hemofilia são causadas por um alelos recessivos

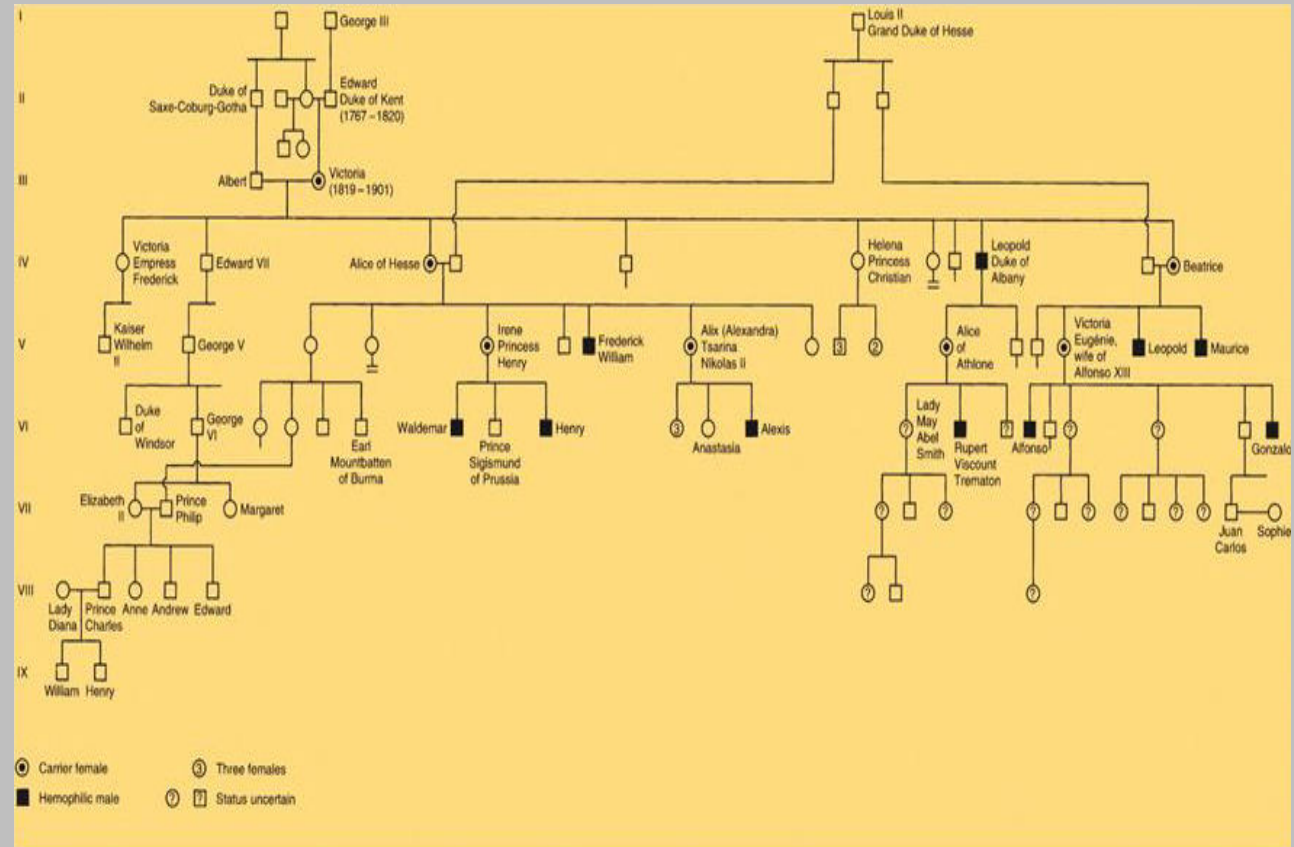


Cystic Fibrosis Lung

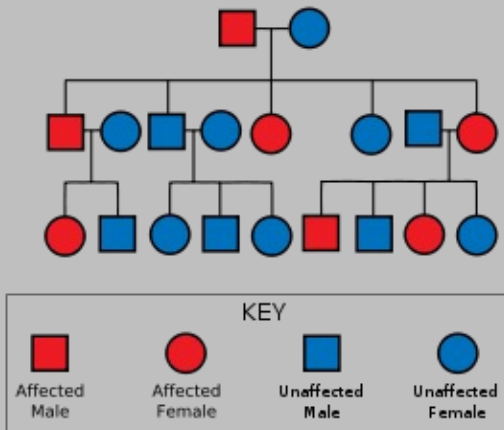
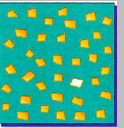
Healthy Lung



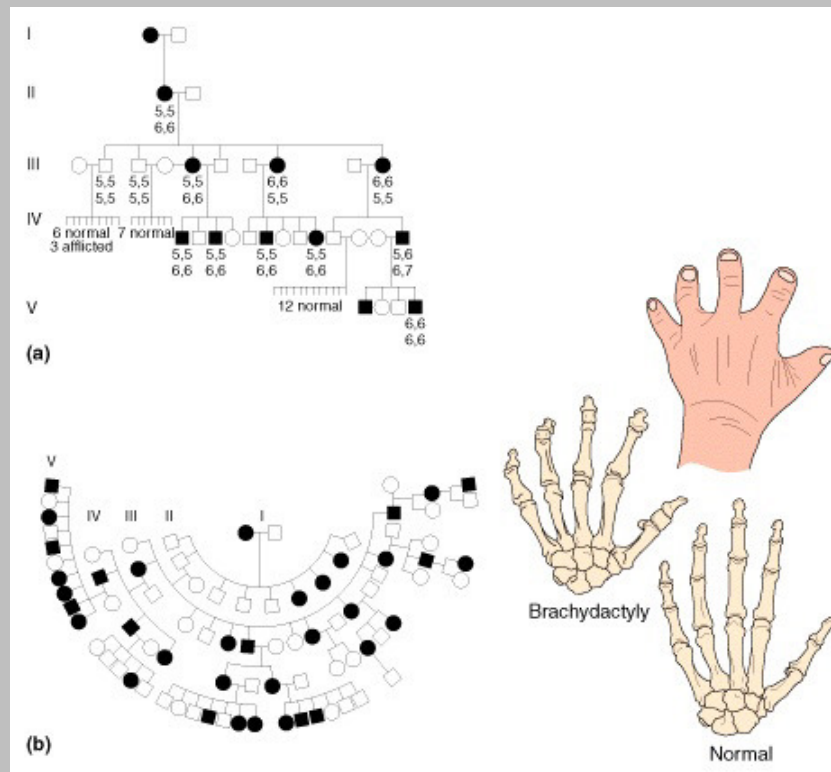
F. Chopin



A doença de Huntington e a polidactilia são causadas por alelos dominantes

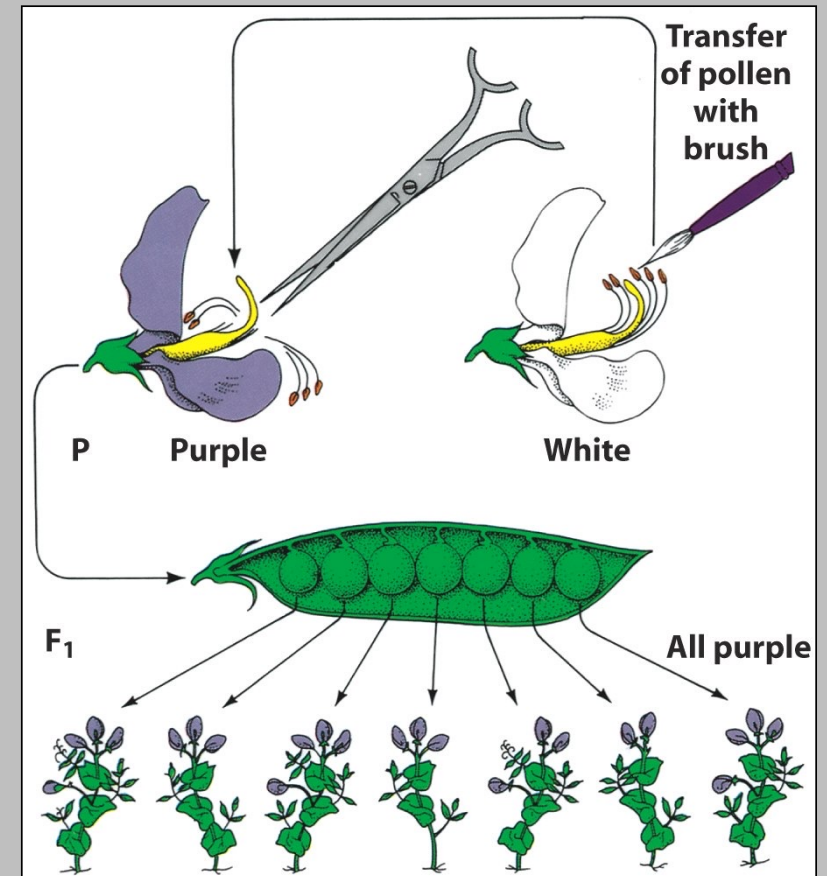
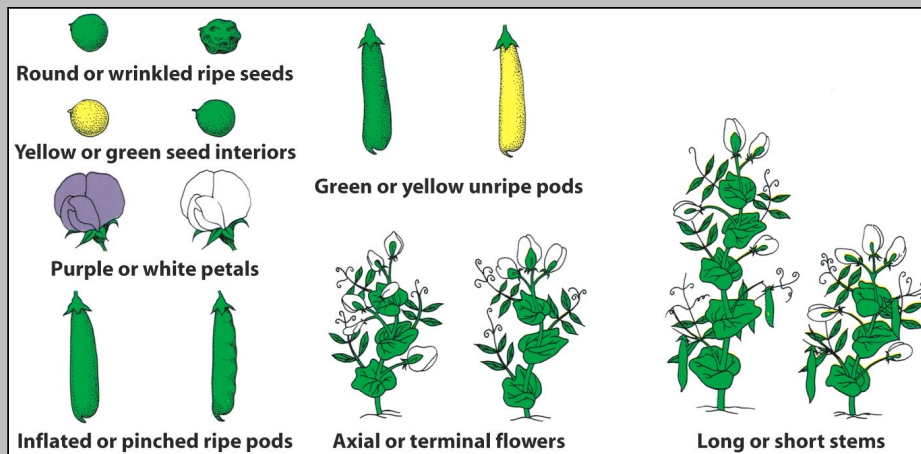


Woody Guthrie

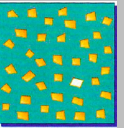


Interacções alélicas

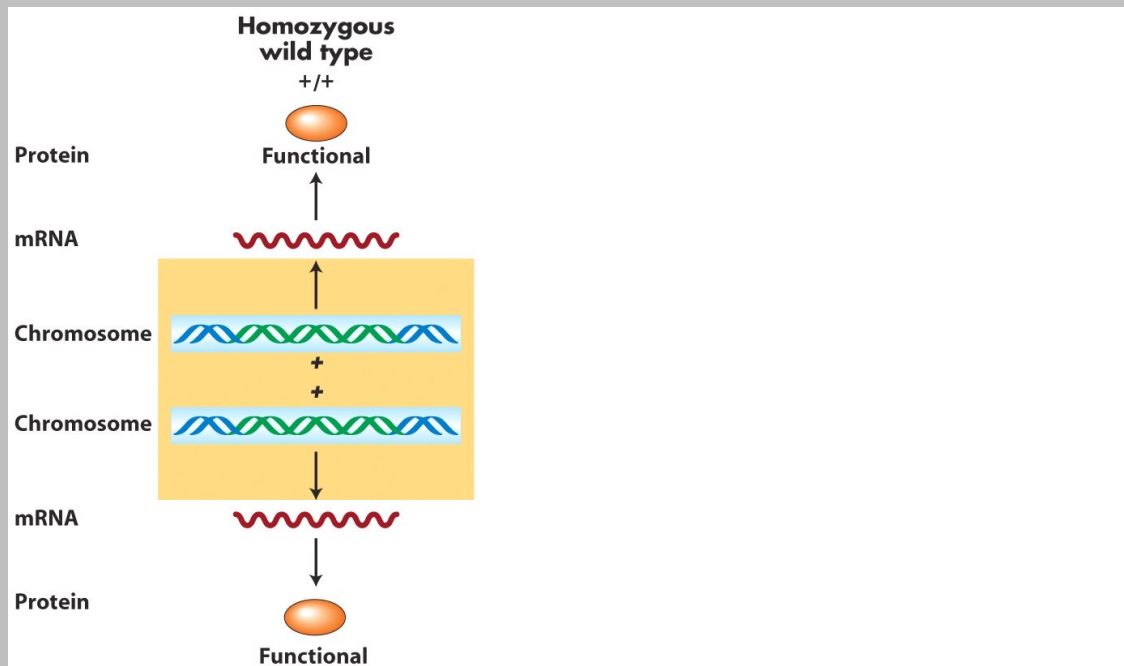
Mendel: Dominância e recessividade



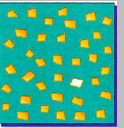
A maioria dos alelos é recessiva



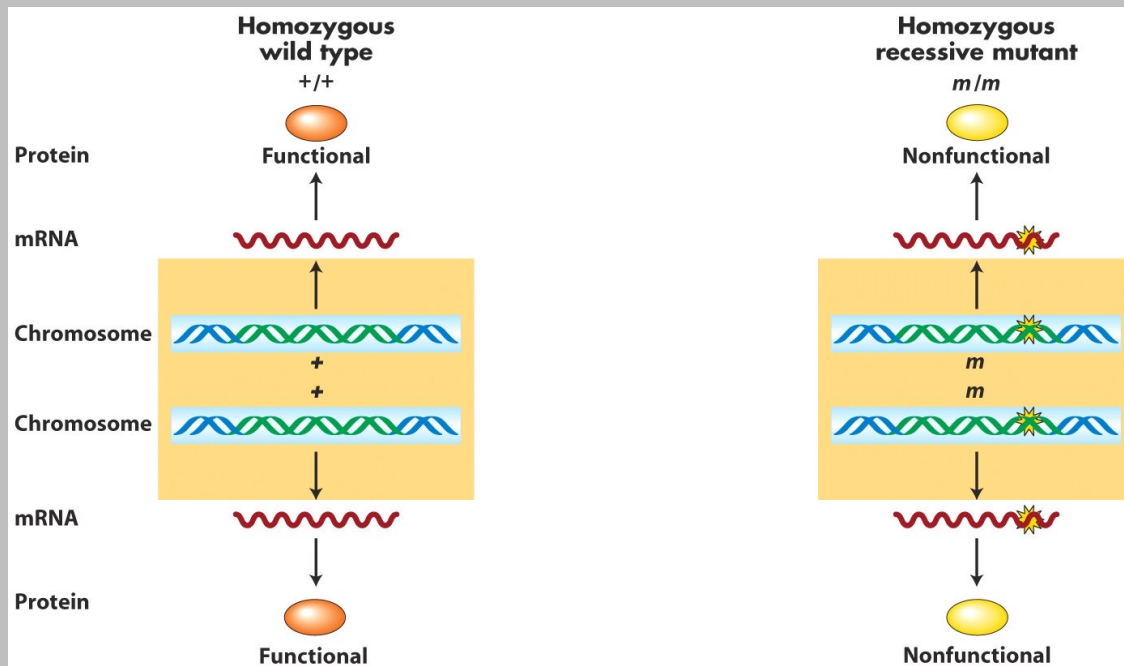
- Os alelos são na sua maioria recessivos, o que implica que uma só cópia funcional é suficiente para o fenótipo selvagem (+).



A maioria dos alelos é recessiva

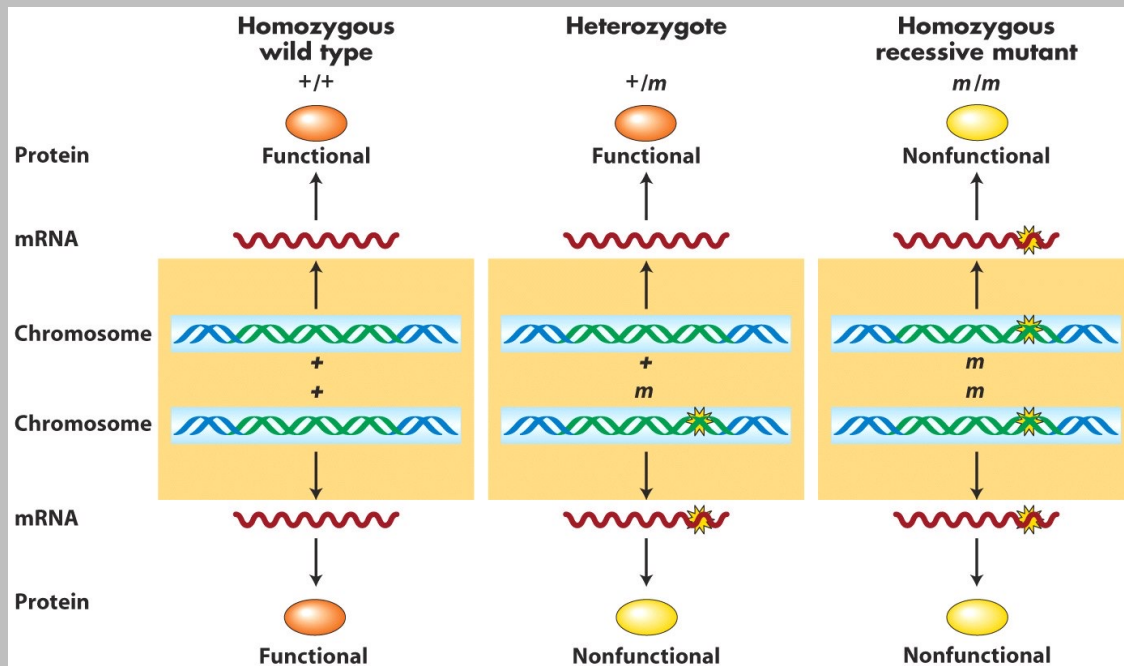


- Os alelos são na sua maioria recessivos, o que implica que uma só cópia funcional é suficiente para o fenótipo selvagem (+).



A maioria dos alelos é recessiva

- Os alelos são na sua maioria recessivos, o que implica que uma só cópia funcional é suficiente para o fenótipo selvagem (+).



Extensões ao Mendelismo

Variações sobre o tema da dominância entre alelos de um mesmo gene

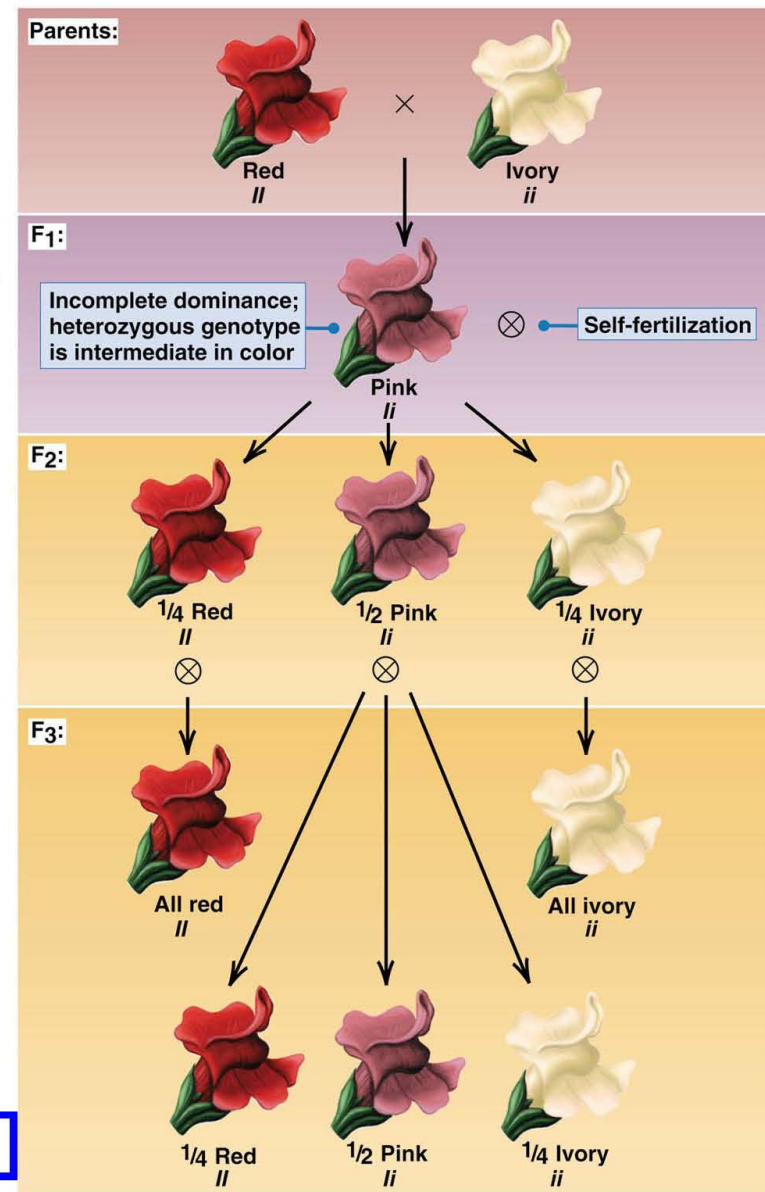
Dominância Incompleta

O fenótipo dos heterozigotos é intermédio do dos homozigotos

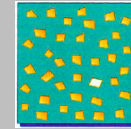
Genes haplo-insuficientes

Nos genes **haplo-suficientes** o fenótipo do híbrido é igual ao do homozigoto dominante.

Antirrhinum majus (bocas-de-lobo)



Dominância incompleta

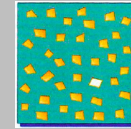


O fenótipo dos heterozigotos é intermédio entre os dois homozigotos numa escala quantitativa.

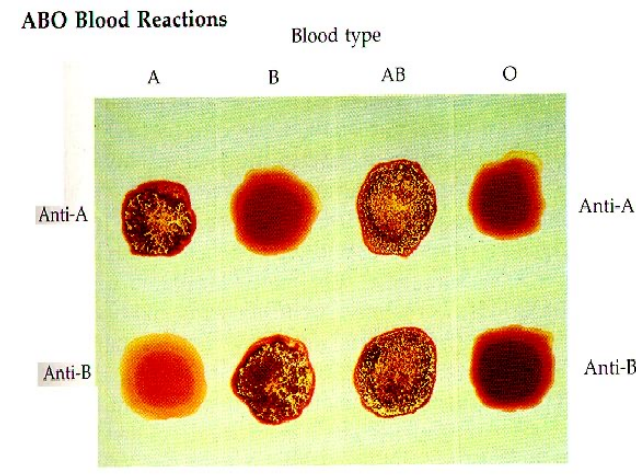
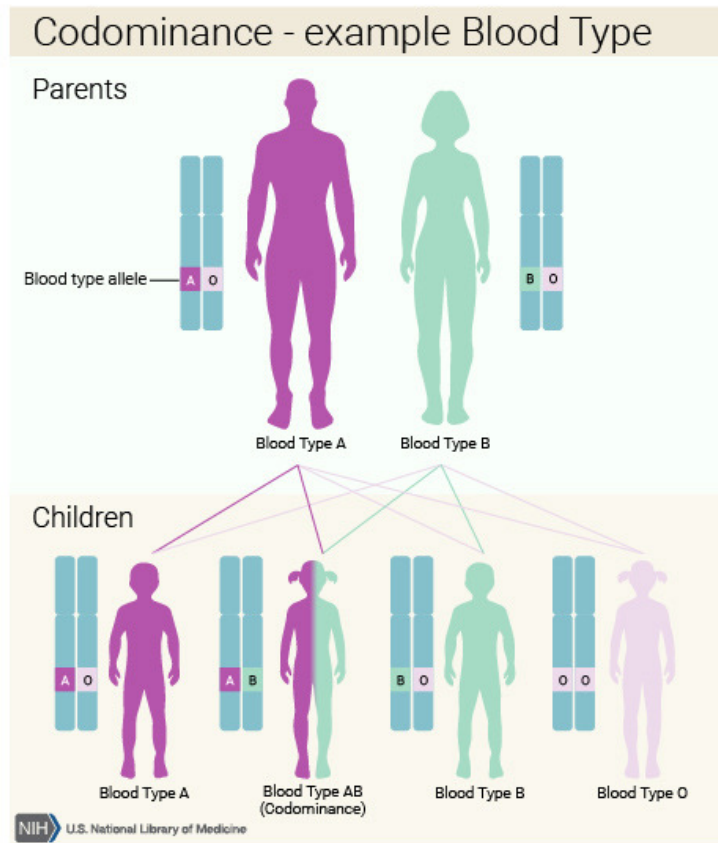
- Imagine uma via de biosíntese de um pigmento. A quantidade de pigmento produzida é função da quantidade de enzima presente na célula.
 - Neste caso, um só alelo funcional produz apenas metade da quantidade de enzima relativamente ao normal e a quantidade de pigmento produzida é metade. Os heteózigóticos têm um fenótipo **intermédio**.
- Four o'clock flowers



Co-dominância



Os heterozigotos expressam o fenótipo de ambos os homozigotos

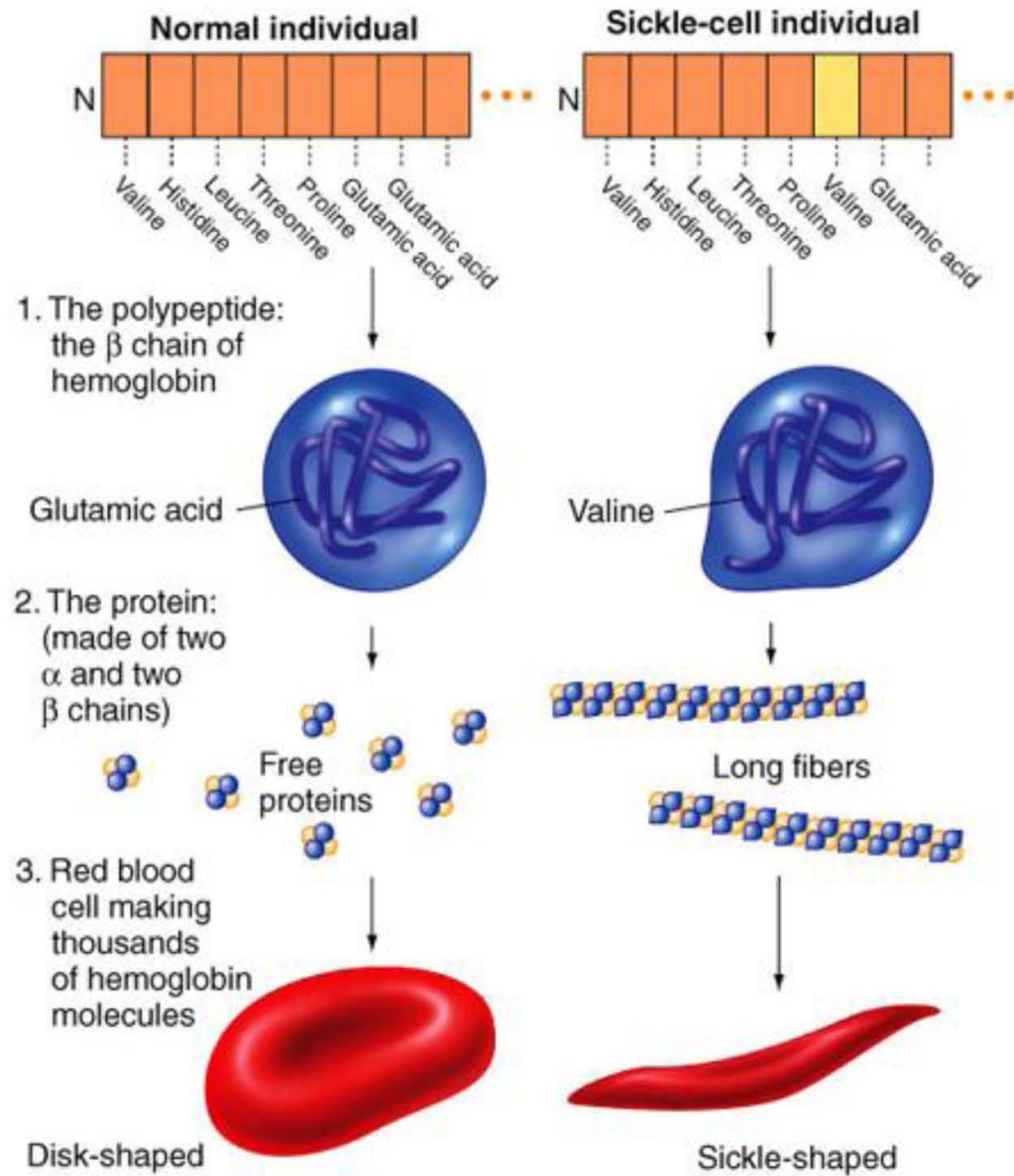


3 alelos I^A , I^B e i

Anemia Falciforme



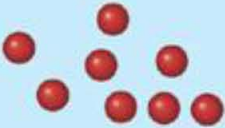
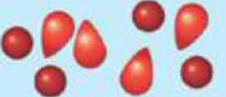
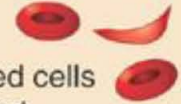
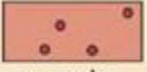
Phenotype	Genotype	Positions to which hemoglobins have migrated		Origin	Hemoglobin types present
		↓	↓	↓	
Sickle-cell trait	Hb^S / Hb^A				S and A
Sickle-cell anemia	Hb^S / Hb^S				S
Normal	Hb^A / Hb^A				A
		←		Migration	



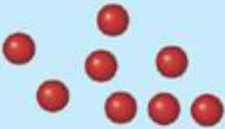
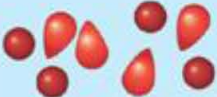
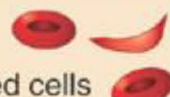
Anemia Falciforme

Mutação na cadeia- β da hemoglobina provoca formação de polímeros anormais de cadeias- α e - β

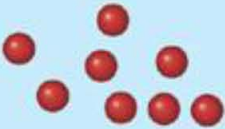
O nível de observação do fenótipo determina a classificação das relações alélicas

Phenotypes at Different Levels of Analysis	Normal <i>Hbβ^A Hbβ^A</i>	Carrier <i>Hbβ^A Hbβ^S</i>	Diseased <i>Hbβ^S Hbβ^S</i>
β-globin polypeptide production			
Red blood cell shape at sea level	Normal 	Normal 	Sickled cells present 
Red blood cell concentration at sea level	Normal 	Normal 	Lower 
Red blood cell shape at high altitudes	Normal 	Sickled cells present 	Severe sickling 
Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia 
Susceptibility to malaria	Normal susceptibility 	Resistant 	Resistant 

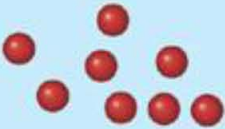
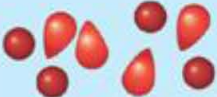
O nível de observação do fenótipo determina a classificação das relações alélicas

Phenotypes at Different Levels of Analysis	Normal <i>Hbβ^A Hbβ^A</i>	Carrier <i>Hbβ^A Hbβ^S</i>	Diseased <i>Hbβ^S Hbβ^S</i>	Dominance Relations at Each Level of Analysis
β-globin polypeptide production				<i>Hbβ^A</i> and <i>Hbβ^S</i> are codominant
Red blood cell shape at sea level	Normal 	Normal 	Sickled cells present 	
Red blood cell concentration at sea level	Normal 	Normal 	Lower 	
Red blood cell shape at high altitudes	Normal 	Sickled cells present 	Severe sickling 	
Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia 	
Susceptibility to malaria	Normal susceptibility 	Resistant 	Resistant 	

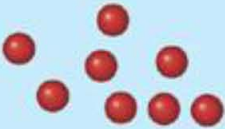
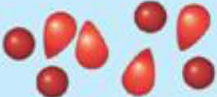
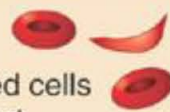
O nível de observação do fenótipo determina a classificação das relações alélicas

Phenotypes at Different Levels of Analysis	Normal <i>Hbβ^A Hbβ^A</i>	Carrier <i>Hbβ^A Hbβ^S</i>	Diseased <i>Hbβ^S Hbβ^S</i>	Dominance Relations at Each Level of Analysis
β-globin polypeptide production				<i>Hbβ^A</i> and <i>Hbβ^S</i> are codominant
Red blood cell shape at sea level				<i>Hbβ^A</i> is dominant <i>Hbβ^S</i> is recessive
Red blood cell concentration at sea level	Normal 	Normal 	Lower 	
Red blood cell shape at high altitudes				
Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia 	
Susceptibility to malaria	 Normal susceptibility	 Resistant	 Resistant	

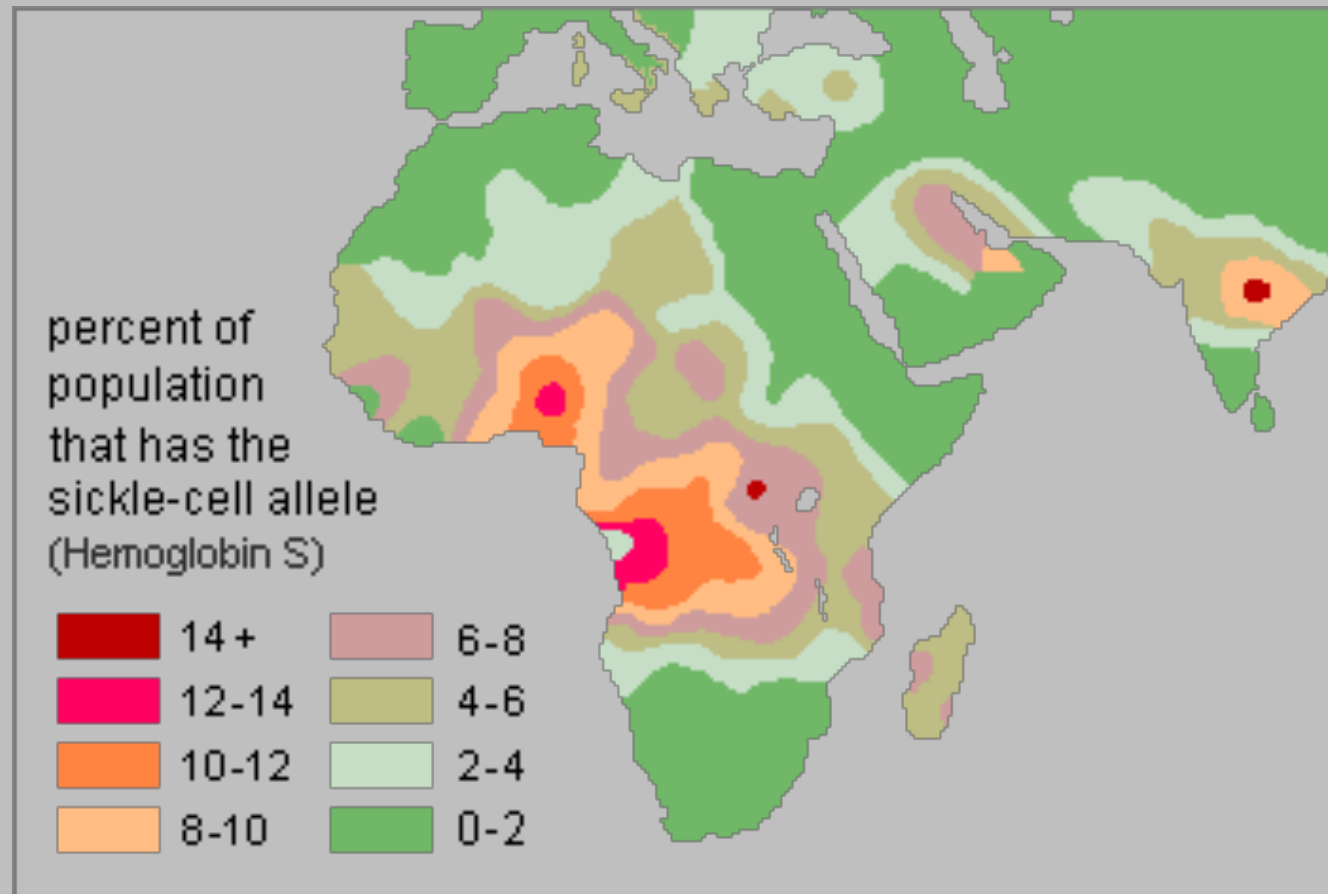
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Red blood cell concentration at sea level	Normal 	Normal 	Lower 	
Red blood cell shape at high altitudes				<i>Hbβ^A</i> and <i>Hbβ^S</i> are codominant
Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia 	<i>Hbβ^A</i> and <i>Hbβ^S</i> show incomplete dominance
Susceptibility to malaria	 Normal susceptibility	 Resistant	 Resistant	

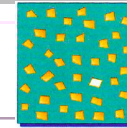
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Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia 	<i>Hbβ^A</i> and <i>Hbβ^S</i> show incomplete dominance
Susceptibility to malaria	Normal susceptibility 	Resistant 	Resistant 	<i>Hbβ^S</i> is dominant <i>Hbβ^A</i> is recessive

Como explica esta distribuição geográfica do alelo S?



Sobre-Dominância



O heterozigoto é mais vigoroso do que os homozigotos

Exemplo: Anemia Falciforme

Frequência elevada nos locais de África onde há malária



A Malaria é provocada pelo protozoário *Plasmodium*

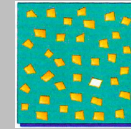
O ciclo de vida deste parasita decorre:

- no interior do mosquito *Anopheles*
- no interior dos glóbulos vermelhos (hemácias)

As hemácias dos heterozigotos $Hb^A Hb^S$ geralmente rebentam quando infectadas pelo *Pasmodium*, impedindo a propagação do parasita

Assim, os indivíduos $Hb^A Hb^S$ têm vantagem relativamente aos:

- ***$Hb^S Hb^S$, porque não sofrem de anemia falciforme***
- ***$Hb^A Hb^A$, porque são mais resistentes à malária***

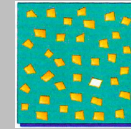


- Cor normal da pelagem dos ratinhos é negra. Uma mutação *Yellow* gera indivíduos de pelagem amarela que quando cruzados com um selvagem geram uma F1 com rácio 1:1.
- No entanto, um cruzamento entre dois ratinhos amarelos resulta num rácio fenotípico 2:1 (2 amarelos para 1 selvagem).

Yellow mice



Letalidade



- A mutação que confere a cor amarela consiste de um alelo do gene ***agouti***. *agouti* controla a deposição de melanina (entre outras coisas).

Yellow mice

